End-to-end utility versus cumulative secure time

CIFAR-10

CIFAR-100

Tiny ImageNet

TFE / ABY3

SecretFlow-SPU

Development-validation accuracy

Cumulative online secure time (s, log scale)

Stage-C detail (8 releases)

Stage-C detail (2 releases)

Stage-C detail (18 releases)

Stage-C online (s)

TFE Full Data

TFE Ours

SPU Full Data

SPU Ours

T@80/90/95

I-bars show sample standard deviation for TFE n=3; Tiny TFE and all SPU panels are n=1. SPU insets show every genuine Stage-C release; backend rows are not hardware-normalized.